O-S-607c
March 20, 1967

SUPERSEDING
Int. Fed. Spec. O-S-00607b (Army-GL)
October 11, 1965 and
Fed. Spec. O-S-607a
July 2, 1953

FEDERAL SPECIFICATION

SODIUM RESINATE, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one grade of sodium resinate (rosin soap) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378—Plastic Film (Polyethylene Thin Gage).

PPP-C-301—Chemicals, Dry and Paste, Packaging and Packing of.

Federal Standards:

Fed. Std. No. 123—Marking For Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Chemical characteristics.** The sodium resinate shall conform to the requirements specified in table I when tested as specified in 4.3.3.

TABLE I. *Chemical characteristics*

Characteristic	Percent by weight	
	Minimum	Maximum
Moisture and volatile matter	—	5.0
Alcohol-insoluble matter	—	1.0
Free alkali, calculated as sodium hydroxide	—	0.10
Free acid, calculated as abietic acid	—	2.0
Matter insoluble in water	—	0.10
Rosin soda soap	90.0	—

3.2 Physical characteristics.

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3.2.1 *Apparent density.* The sodium resinate shall have an apparent density of between 23 and 30 pounds per cubic foot when tested as specified in 4.3.3.

3.2.2 *Color.* The sodium resinate shall have a uniform color not darker than color No. 30266 of Fed. Std. No. 595.

3.2.3 *Appearance.* The sodium resinate shall be a uniform, noncaking, free-flowing powder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 *Component and material inspection.* In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 *Examination of sodium resinate.* The sodium resinate shall be examined for the defects listed herein. The inspection level shall be S-4 and the acceptable quality level (AQL) shall be 2.5, expressed in defects per hundred units. The lot size shall be expressed in units of unit containers. The sample unit shall be one filled unit container.

Examine	Defect
Appearance	Not uniform. Caking Not a free-flowing powder.
Color	Color not uniform. Darker than color chip No. 30266

4.3.2 *Examination of preparation for delivery.* The packaging, packing, and marking requirements of this specification shall be examined in accordance with the provisions of PPP-C-301, except that the inspection level shall be S-2 and the AQL shall be 2.5, expressed in defects per hundred units.

4.3.3 *Testing of the sodium resinate.* The methods of testing specified in Fed. Test Method Std. No. 536 wherever applicable and as listed in table II shall be followed for each lot. The lot size shall be expressed in pounds of sodium resinate. The sample unit shall be 1-1/4 pounds of sodium resinate. The sample size shall be as specified below. As specified in table II, tests shall be performed on the composite or on the sample units. One pound of sodium resinate shall be taken from each sample unit and thoroughly mixed to obtain the composite. Each sample unit and the composite shall be placed in separate sealed airtight and watertight metal or glass containers. The two specimens from each sample unit, for moisture and volatile matter determination, shall be weighed immediately upon opening the sealed sample unit containers. Care shall be exercised to prevent the loss or absorption of moisture during the sampling, compositing, storage, and weighing of the specimens and samples. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if any sample unit or the composite fails to meet any test requirement specified.

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE II. *Instructions for testing of the end item*

Characteristic	Specification reference		Requirements applicable to		Number of determinations per unit	Results reported as
	Requirement	Test method	Samp. unit	Comp. unit		Numerically to nearest
Moisture and volatile matter (oven method)	Table I	201	X		2	Average of 2 determinations to nearest 0.1 percent.
Alcohol-insoluble matter	Table I	301		X	2	Average of 2 determinations to nearest 0.1 percent.
Free alkali, calculated as sodium hydroxide	Table I	302		X	2	Average of 2 determinations to nearest 0.1 percent.
Free acid, calculated as abietic acid	Table I	302[1]		X	2	Average of 2 determinations to nearest 0.1 percent.
Matter insoluble in water	Table I	303		X	2	Average of 2 determinations to nearest 0.01 percent.
Rosin soda soap	Table I	701		X	2	Average of 2 determinations to nearest 0.1 percent.
Apparent density	3.2	4.4.1		X	2	Average of 2 determinations to nearest [illegible] per cu. ft.

[1] To express free acid as abietic acid, substitute 30.24 for 28.25 as factor for the calculation.

4.4 Test methods.

4.4.1 *Apparent density.* Fill a 250-ml. weighed graduated cylinder to approximately the 240 ml. mark with the sample. Settle the material by raising the graduate approximately 1 inch above the surface of a piece of lead sheet on a flat surface and allow it to fall of its own weight. Repeat until a total of four falls have been given. Read the volume of the settled material. Weigh the graduate and contents to the nearest 1.0 g. Calculate as follows:

$$\text{Apparent density (lbs./cu. ft.)} = \frac{\text{Wt. of sodium resinate in g.} \times 62.43}{\text{Volume in ml. occupied after settling}}$$

5. PREPARATION FOR DELIVERY

5.1 **Level A, B, and C.** Fifty pounds of the material shall be packaged and packed for each level of protection in a fiberboard drum in accordance with type IV, class 1 of PPP-C-301, with a liner fabricated from material conforming to L-P-378.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special markings required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The sodium resinate covered by this specification is intended for use as an ingredient of cleaning or degreasing compounds which are used to clean metal parts.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Sodium Resinate (Rosin Soap) - O-S-607c

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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